MANOJ KUMAR. C

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PROFILE SUMMARY

Engineering graduate specializing in embedded systems and firmware engineering with published research and hands-on industry experience. Proven expertise in Android Application Development, BIOS porting in Linux, microcontroller programming, and collaborative software development. Former IEEE SSCS Chapter Chairperson with certifications in Generative AI and Prompt Engineering.

AREAS OF INTEREST

- Operating System & Firmware Engineering
- Simulating Electrical Circuit Analysis
- Embedded programming (Microcontrollers and Processors).
- Version Control and Collaborative Software Development
- Machine Learning and IoT for Electrical machines
- Space Exploration (Moon Landings, Space Shuttles)

EDUCATION

- B.E. Electrical and Electronic Engineering
  Sri Sairam Engineering College, Graduation: 2025
  CGPA: **9.09** / 10

School Education:

- Higher Secondary (12th): **93.02%**
- Secondary (10th): **88.4%**

WORK EXPERIENCE

Senior Software Engineer – OS Team – Aug 2025 to Oct 2025

HCL Technologies (Xerox Project) | 2 Months

- Involved in defect consulting for upcoming software releases within OS team.
- Collaborated using Git workflows (branching, merging, pull requests, code integration)

Android Development – Oct 2025 to Present

HCL Technologies (Xerox Project) | Present

- Involved in bug fixes and Feature implementation in Xerox Kiosk Workflows.
- Collaborated using Git workflows (branching, merging, pull requests, code integration)

INTERNSHIPS

Industry Intern

HCL Technologies | 6 months

- Executed industry project: "*Porting of Open-Source Intel BIOS to device-specific firmware*".
- Gained expertise in BIOS configuration and low-level firmware customization.

Embedded Systems Intern

National Small Scale Industries Technical Service Centre, Guindy Industrial Estate | 2 weeks

Power Electronics and IoT Intern

IndFurr SuperHeat Furnaces, Ambattur Industrial Estate | 5 weeks

SKILLS & ABILITIES

- Programming: C, C++, Python
- Development Tools: Android Studio, Visual Studio Code, Platform IO
- Version Control: Git
- CAD Software: Proteus Professional, DIALux Evo

- FPGA Tools: Xilinx Vivado & Active HDL (Beginner)
- PCB Designing: Autodesk Eagle, PCB123
- Systems: Linux, Shell Scripting, Terminal Utilities
- Automation: UiPath Studio

PROFESSIONAL DEVELOPMENT

Leadership:

- Chairperson – IEEE Solid-State Circuits Society Student Branch Chapter (2022 – 2024)

Conferences and Publications:

- Presented our Project paper on "Wireless Servosystem for AC and DC Machines" in IEEE Yesist'12, 2023 Conference.
- Published our Project paper "Wireless Servosystem for AC and DC Machines" in International Conference on Power, Energy, Control and Transmission Systems (ICPECTS 2024).
- Co-authored research on project "Cascaded Neural Network Controller Development for Electric Vehicle Energy Management" in ICPECTS 2024.

Certifications:

- C Training – IIT Bombay (Spoken Tutorial)
- IoT Foundations – LinkedIn Learning
- Python for Machine Learning – Great Learning Academy
- UiPath Studio – UiPath Academic Alliance
- Generative AI & Prompt Engineering – Various Industry Courses

Workshops Attended:

- *Robotic Process Automation (RPA)* - UiPath Academic Alliance
- *Android Development* - Panimalar Engineering College
- *Introductory Course on Universal Human Values* - AICTE
- *Product Development and Entrepreneurship Workshop* - SRM Engineering College
- *VHDL Programming for FPGA* - Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College

PROJECT

Wireless Servosystem for AC and DC Machines (Final Year Project – Outstanding Grade)

- Designed an embedded servo system with Capacitive Touch for wireless control of AC and DC machines.
- Real-time analysis of machine characteristics for comprehensive communication with the operator.
- Utilized *FPGA* and *VHDL* programming for real-time working and analysis.
- Used *Lora Tech* for Communication protocol promoting industrial automation and control through the use of advanced tech.